
ALZHEIMERNET: MULTIMODAL ALZHEIMER DISEASE CLASSIFIER WITH DUAL EXPLAINABILITY

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ABSTRACT

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder, marked by irreversible cognitive decline. Mild Cognitive Impairment (MCI) is an intermediate stage that often precedes AD, making early and accurate diagnosis essential for timely intervention.

Achieving this goal, however, remains challenging. Most high-performing models operate as "black boxes," offering no insight into why a prediction is made, and distinguishing cognitively normal (CN) subjects from MCI remains difficult due to subtle anatomical differences and high inter-subject variability.

We propose **AlzheimerNet**, an interpretable framework for Alzheimer’s disease classification that combines a supervised **2.5D ResNet-18** with a reconstruction-based approach using three class-specific multimodal autoencoders (CN, MCI, AD). The framework provides dual explainability through **Grad-CAM** saliency maps and reconstruction-based anomaly maps.

Evaluated on the **ADNI4** dataset using structural MRI modalities (T1, T2, and FLAIR), the reconstruction-based classifier achieved a weighted F1-score of **0.52**, while the supervised classifier obtained a balanced accuracy of **0.468** (AUC **0.724**) for the three-class task. Binary classification performance reached a balanced accuracy of **0.690** (AUC **0.687**) for CN vs. MCI and **0.650** (AUC **0.828**) for CN vs. AD.

Overall, the results demonstrate that combining accurate classification with interpretable visual explanations provides a promising decision-support framework for Alzheimer’s diagnosis, while confirming that distinguishing **MCI** remains the most challenging task due to its transitional nature.

1 Introduction

Alzheimer’s Disease (AD) is a progressive neurodegenerative disorder typically preceded by Mild Cognitive Impairment (MCI), an intermediate stage marked by measurable cognitive decline. Accurate discrimination between cognitively normal (CN), MCI, and AD subjects is essential for early diagnosis and timely intervention.

This task remains challenging due to subtle and heterogeneous brain changes in early disease stages. While biomarkers such as hippocampal atrophy, cortical thinning, and white-matter alterations are informative, they are not sufficient alone to reliably separate adjacent stages [17].

Deep learning, particularly convolutional neural networks (CNN), has shown strong ability to learn discriminative patterns from structural MRI. Motivated by this, it is proposed a model to classify subjects along the AD continuum (CN, MCI, AD) as both a multi-class and a set of binary tasks (CN - AD, CN - MCI, AD - MCI).

The system produces probabilistic, patient-level predictions by integrating information across multiple MRI modalities (T1, T2, FLAIR), better reflecting clinical diagnostic practice.

Beyond classification performance, interpretability is a key requirement due to the high-stakes nature of clinical decision-making. The system is designed to highlight the brain regions most influential for each prediction using Grad-CAM, allowing clinicians to verify whether decisions are consistent with known neuropathological patterns rather than artefacts or scanner-specific biases.

This is further reinforced through a reconstruction-based strategy using class-specific autoencoders. Each model learns a compact representation of CN, MCI, and AD brain patterns across T1, T2, and FLAIR modalities, integrating their complementary structural information. For a given scan, reconstruction quality indicates similarity to each class, while poorly reconstructed regions identify anatomical deviations and generate spatial anomaly maps.

Together, these complementary approaches aim to provide clinically interpretable evidence. The proposed system is evaluated on the ADNI-4 dataset, a large multi-site benchmark for Alzheimer’s disease imaging studies.

2 Related Work

Deep learning approaches to Alzheimer’s disease detection from structural MRI have been dominated by convolutional network architectures (CNN), although other approaches were also considered in the literature to address limited sized datasets such as ADNI [1].

CNN-based classification. CNN have been extensively benchmarked across AD datasets and consistently identify *hippocampal and medial temporal lobe atrophy* as the most discriminative regions [2]. More recent hybrid designs combine a ResNet or VGG backbone with Vision Transformer (ViT) self-attention to jointly capture local structural features and long-range connectivity patterns [3]. These architectures achieve state-of-the-art accuracy on curated benchmarks, but typically *operate on a single MRI modality* (T1-weighted) and rely on large or pre-trained models.

Graph Neural Networks (GNNs). A second direction models the brain as a graph whose *nodes correspond to anatomical regions* of interest derived from a brain atlas and whose edges encode structural or functional relationships; GNNs are then applied for classification [4]. This paradigm offers anatomically interpretable representations, but its performance is critically *sensitive to the choice of graph topology* that requires neuroimaging expertise not always available within a team [5].

Multimodal MRI fusion. Combining complementary MRI sequences (T1, T2, FLAIR) consistently improves diagnostic performance over any single modality. Existing fusion strategies range from early concatenation of input channels to late fusion of modality-specific features and attention-based weighting [6, 7]. Despite this evidence, most published architectures fuse at most two modalities and apply a single, shared encoder to all of them, *discarding any potential for modality-specific or class-specific latent specialisation*.

Autoencoder(AE)-based anomaly detection. Reconstruction-based unsupervised anomaly detection trains a model on healthy anatomy and flags deviations in pathological scans as high reconstruction error [8]. UNet variants, which introduce skip connections to preserve high-frequency spatial detail during reconstruction, produce sharper and more localised error maps than vanilla-AE [9]. Applications to AD, however, remain inherently binary (healthy vs. anomalous): a single autoencoder trained on CN subjects cannot distinguish MCI from AD, since both classes deviate from the CN prototype by construction.

Class-specific autoencoders. The idea of training one AE per class and classifying an unseen sample by assigning it to the class whose reconstructor achieves the lowest error has been formalised in the computer-vision literature. Bojanowski and Joulin [10] apply a stack of discriminative class-specific AEs to multiclass anomaly detection in medical endoscopy images, using a combined reconstruction and inter-class distance loss.

No prior work appears to have applied this paradigm to multimodal neuroimaging for ternary AD staging, nor combined it with a parallel classification branch and dual explainability mechanisms.

Explainability in neuroimaging. Grad-CAM and its variants remain the dominant explainability tool for CNN-based models, producing spatial heatmaps that attribute predictions to image regions [11]. For graph-based models, GNNExplainer serves an analogous role by identifying influential subgraphs [18]. Recent surveys on explainable AI in dementia emphasise that interpretability methods are most clinically valuable when *multiple independent explanation signals can be cross-validated against one another*, rather than relying on a single tool in isolation [5].

The present work follows this recommendation by combining reconstruction-based anomaly maps with Grad-CAM heatmaps.

3 Materials

The experiments were conducted on the ADNI4 dataset (Figure 1), comprising structural brain MRI from 817 subjects. Each subject underwent one or more scanning sessions, resulting in 952 sessions in total. In particular, 122 subjects were scanned more than once, at screening (sc), baseline (init), 12-month (m-12), or 24-month (m-24) timepoints. Every session produced three co-registered volumes: a T1-weighted, a T2-weighted, and a FLAIR scan, giving 2,856 volumes overall (952 per modality). For each session, the dataset further provides the associated demographic and clinical metadata, namely the subject’s age and sex, the diagnostic label, and the visit timepoint.

Based on clinical assessment, subjects were divided into three diagnostic groups: 466 cognitively normal (CN), 295 with mild cognitive impairment (MCI), and 56 with Alzheimer’s disease (AD), corresponding to 490, 389, and 73 sessions respectively.

The dataset is partitioned into training, evaluation, and test subsets, comprising 664 (69.7%), 141 (14.8%), and 147 (15.4%) sessions, respectively.

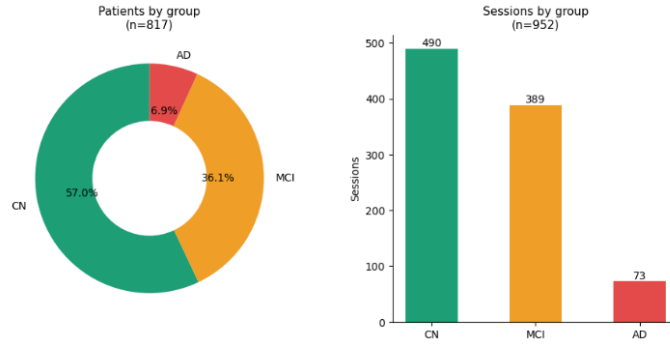


Figure 1: ADNI4 dataset analysis

3.1 Data Pre-Processing

Prior to training, all MRI volumes underwent a standardized preprocessing pipeline designed to improve data consistency while preserving anatomical information.

The structural MRI volumes underwent a standardized preprocessing pipeline designed to reduce inter-subject variability, remove irrelevant anatomical regions, and produce inputs with consistent dimensions suitable for deep learning.

Intensity Clipping: Each MRI volume was loaded in NIFTI format using the NiBabel library. Since MRI intensities should not contain meaningful negative values, all voxel (3D pixel) intensities below zero were clipped:

$$I(x, y, z) = \max(I(x, y, z), 0)$$

Brain Region Cropping: For each axial slice, the amount of brain tissue was estimated by counting voxels above an intensity threshold, and slices whose tissue area fell below a fraction of the maximum observed area were discarded. This procedure removes uninformative slices located near the skull base, as well as near the vertex of the skull, while preserving the central region containing structures such as the hippocampus and lateral ventricles, which are particularly relevant for Alzheimer’s-related classification.

Cropping was restricted to the z-axis only, leaving the x and y axes unmodified, in order to avoid introducing spatial bias or excessive geometric distortion on multiple axes.

Volume Resampling: Volumes were resampled to a fixed resolution using trilinear interpolation. For original dimensions (H, W, D) and target size $(H_t = 128, W_t = 128, D_t = 128)$:

$$s_x = \frac{H_t}{H}, \quad s_y = \frac{W_t}{W}, \quad s_z = \frac{D_t}{D}.$$

All modalities were resized to identical spatial dimensions, ensuring consistency across subjects.

Intensity Normalization: Voxel intensities were normalized per volume:

$$I_{norm} = \frac{I}{I_{max} + \varepsilon}, \quad \varepsilon = 10^{-8}$$

resulting in values approximately in $[0, 1]$, improving training stability while preserving contrast.

Contrast Enhancement: CLAHE was applied slice-wise to improve local contrast. For each slice, a foreground mask was defined using $I_{norm} > 0.1$. CLAHE (clip limit 0.01) was applied only if the foreground contained more than 100 voxels, and only foreground voxels were updated, preserving background structure.

Slice Selection: A fixed number $N = 16$ of axial slices was uniformly sampled, rather than using the full set of 128 slices. Slices were selected at evenly spaced intervals starting from a fixed axial offset ($Start_{slice} = 55$), determined empirically by visual inspection as the point where cerebral tissue first appears, below which the volume contains only skull and bone structures. This choice ensures that no structural brain information is discarded by an overly conservative offset, while excluding clearly non-cerebral regions. Due to natural inter-subject anatomical variability, the exact tissue content at this offset is not perfectly aligned across patients; a small amount of residual cerebellar tissue may therefore still be present in the lowest sampled slices for some subjects. This is an accepted limitation of the fixed-index sampling strategy, affecting only a minority of the sampled slices, and represents a deliberate trade-off against the added complexity of anatomical atlas-based alignment.

Slice Resizing: Each selected slice was resized to the network input resolution using bilinear interpolation, ensuring uniform spatial dimensions across all samples.

Multi-Modal Alignment: Each sample consists of three aligned channels (T1, T2, FLAIR). All geometric operations were applied consistently across modalities, preserving voxel-wise correspondence and forming a three-channel input.

Data Augmentation To address class imbalance, online augmentation was applied to AD subjects only. Deterministic transformations included horizontal flipping ($p=0.5$) and in-plane rotations in $[-5^\circ, 5^\circ]$, applied consistently across modalities using a subject-specific seed.

4 Methodology

4.1 Reconstruction-based Classifier

Three independent multimodal autoencoders, one per clinical class (CN, MCI, AD). Each class-specific autoencoder is itself composed of three modality-specific encoders (T1, T2, FLAIR), whose embeddings are aggregated into a single fused embedding for that class, which is then decoded by a single class-specific decoder. Each 3D MRI volume is processed as a sequence of independent 2D axial slices rather than as a full 3D volume.

The transformation from volumetric data to slice-wise processing is illustrated in Figure 2.

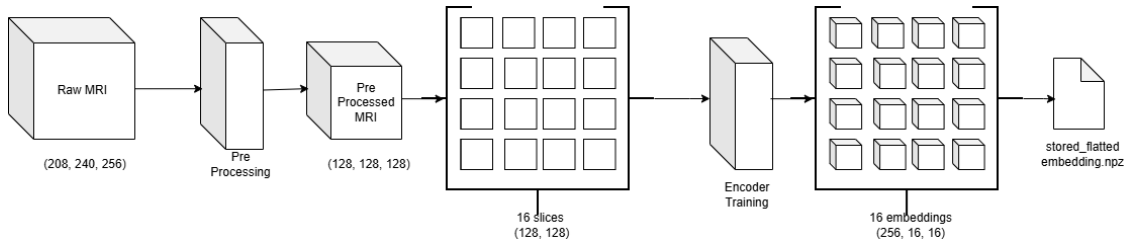


Figure 2: From 3D volumes to 2D slice-wise representations

To preserve information from all imaging modalities while maintaining their distinctiveness, encoder outputs are fused via **channel-wise concatenation**, a lossless operation that preserves modality-specific features in fixed positions within the representation. The resulting slice-level embeddings are then aggregated across the axial dimension using a **Gaussian-weighted mean**, which emphasizes central slices that are typically more anatomically informative:

$$w_k \propto \exp\left(-\frac{z_k^2}{2\sigma^2}\right), \quad z_k \in [-1, 1],$$

In total, the architecture consists of nine encoders and three decoders.

The overall autoencoder framework is illustrated in Figure 3.

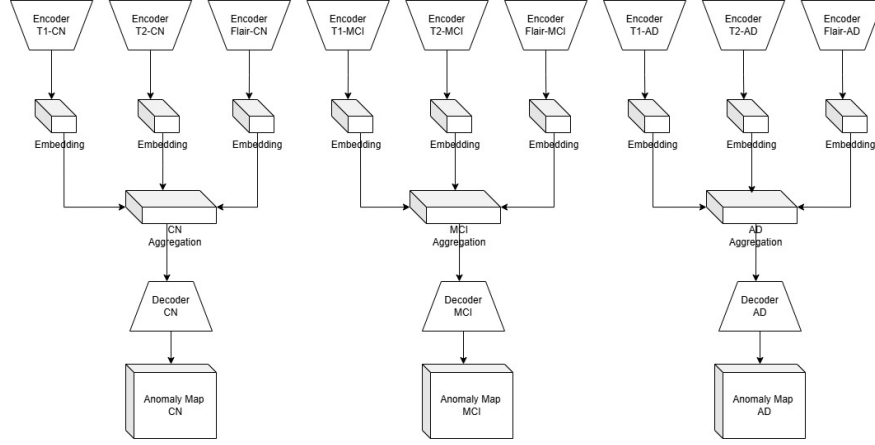


Figure 3: Overall architecture of the class-specific multimodal autoencoders

The encoder and decoder components are further detailed in Figures 4 and 5, respectively.



Figure 4: Encoder architecture details

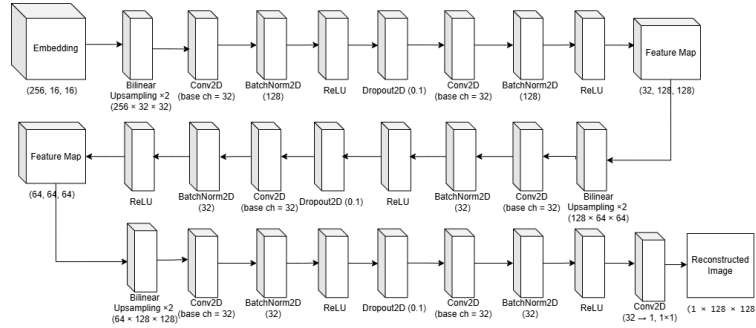


Figure 5: Decoder architecture details

For each class-specific autoencoder, reconstruction errors are computed independently, producing three anomaly maps per subject (CN, MCI, AD) for each of the 16 selected slices. The predicted class is the one with the lowest reconstruction error, reflecting the closest anatomical match. Because the three decoders are trained on substantially imbalanced data (49 AD sessions versus 343 CN and 272 MCI sessions), their raw reconstruction losses are not directly comparable and we need to apply a **bias calibration**. Slice-wise reconstruction losses are aggregated as

$$L_{\text{raw}}^{(g)} = \sum_{k=1}^{16} w_k \ell_k, \quad L_{\text{calibrated}}^{(g)} = \frac{L_{\text{raw}}^{(g)}}{B_g}, \quad \hat{g} = \arg \min_g L_{\text{calibrated}}^{(g)}.$$

A lower calibrated loss indicates that the decoder trained on class g reconstructs the input more faithfully, yielding the predicted diagnosis.

4.2 Hybrid CNN-based Classifier Architecture

The proposed framework leverages a CNN-based approach, motivated by the limited size of the ADNI dataset and the strong inductive bias of convolutional networks, which improves generalization in small-data regimes and enables native compatibility with Grad-CAM for interpretability.

An initial classifier was built on top of pretrained multimodal autoencoder encoders, where modality-specific embeddings (T1, T2, FLAIR) of shape (256, 16, 16) were aggregated, then reduced to a 256-dimensional vector through Global Average Pooling (GAP) and finally projected into a shared latent space.

These representations were then fused using **attention pooling**, which is employed to adaptively weight each modality according to its relevance for the current subject, rather than assuming equal contribution. This is particularly important in neuroimaging, where the diagnostic signal may be more pronounced in one modality depending on the disease stage.

You can observe the detailed architecture in the representation below (Figure 6).

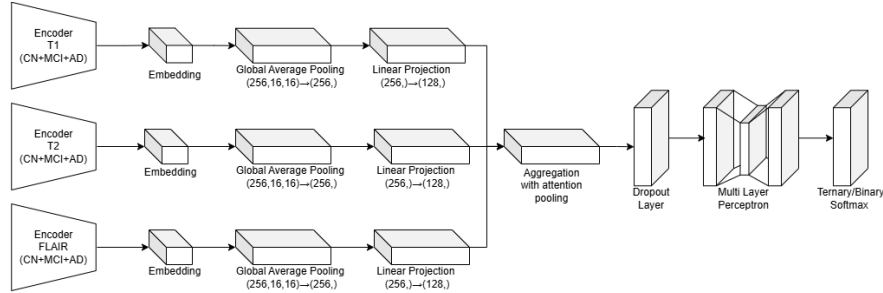


Figure 6: Architecture of the Classification v.1.0

Despite this design, the resulting representations proved insufficiently discriminative, leading to a **majority-class collapse across all tasks** (AUC = 0.500), with performance metrics aligning with class distribution priors. This indicated that reconstruction-trained embeddings did not capture enough task-specific structure for classification.

These findings motivated the adoption of a dedicated ResNet-18 backbone trained end-to-end for supervised classification, using the same multimodal input but optimized directly for discriminative learning rather than reconstruction-based representations.

4.3 ResNet-18 Based Classifier

At the core of the proposed approach we have a ResNet-18 CNN for classifying Alzheimer’s disease stages from structural MRI. The model is initialized via transfer learning and fine-tuned end-to-end, with a lightweight classification head composed of dropout and a final linear layer sized to the number of target classes.

T1, T2, and FLAIR modalities are combined into a single multi-channel input to jointly exploit complementary structural information (e.g. gray-matter atrophy and white-matter abnormalities). Although training is performed at the slice level, predictions are aggregated at the patient level, which is the unit of actual clinical interest, by averaging slice-wise softmax probabilities, yielding a single set of class probabilities for that patient. The predicted class is then the one with the highest averaged probability, so that no individual slice is allowed to dominate the decision and the final prediction instead reflects the consensus of evidence gathered across the whole scan.

Class imbalance, arising from the uneven distribution of training sessions (343 CN, 272 MCI, 49 AD), is addressed through a combination of sampling, loss reweighting, and targeted augmentation. Specifically, a `WeightedRandomSampler` is used to balance class occurrence at the batch level, while `inverse-frequency` class weights are incorporated into a label-smoothed cross-entropy loss ($\epsilon = 0.1$) to further mitigate bias during optimization.

At inference time, per-slice softmax probabilities are averaged across all slices belonging to the same session (with augmented copies merged back into their parent session via session-ID normalization) to yield a single patient-level prediction, aligning the evaluation protocol with the clinically meaningful unit of analysis — the patient, not the slice.

The architecture of this classifier is represented in the following figure:

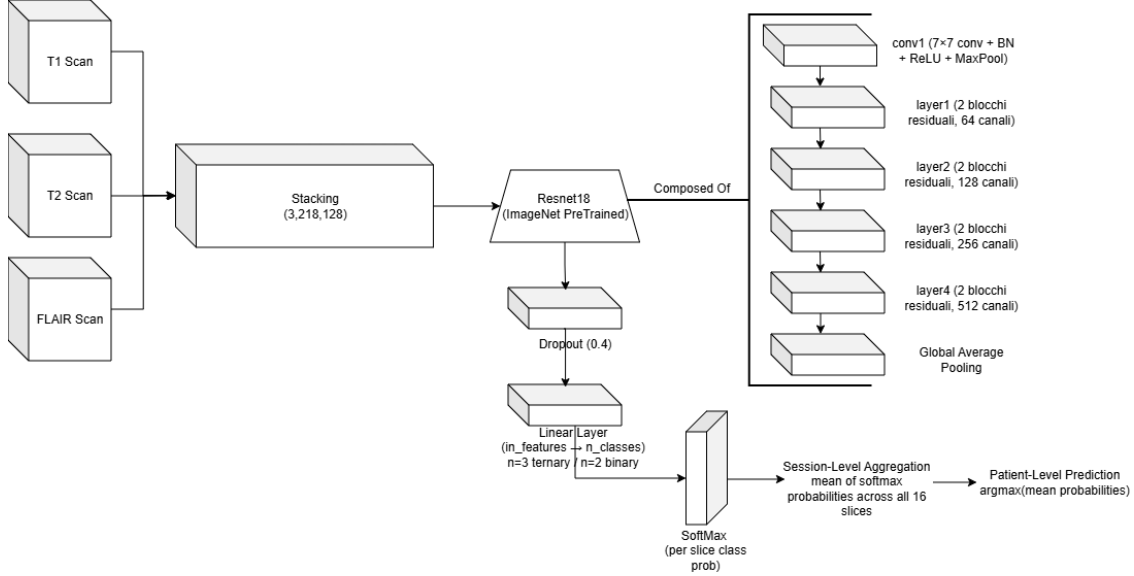


Figure 7: Architecture of the ResNet18 Classifier

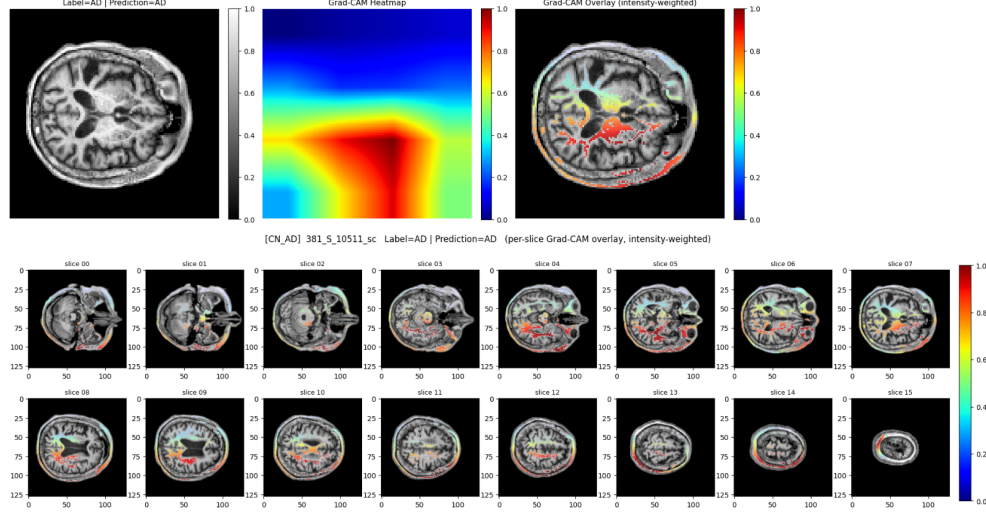
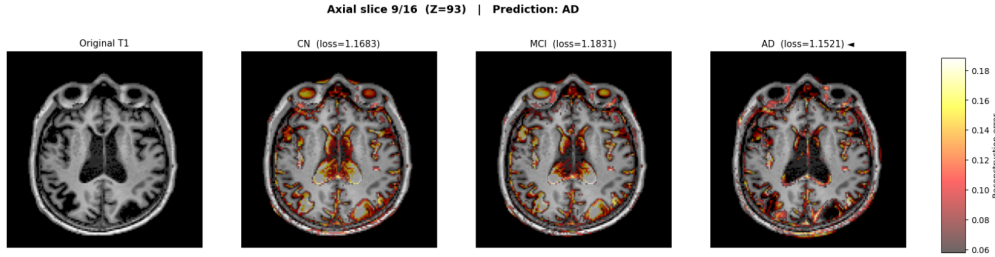
4.4 Interpretability

Interpretability is treated as an integral requirement of the framework rather than a diagnostic afterthought, and is addressed through two complementary mechanisms operating at different levels of the pipeline.

The first is Grad-CAM, applied to the final convolutional layer of the classifier to produce, for each slice, a spatial map of the image regions that most influenced the predicted class. Since a subject-level explanation is preferable to a disconnected set of per-slice maps, these individual saliency maps are also combined into a single fused map per patient through a weighted average that favors the more central slices, mirroring the same anatomical relevance criterion used when the representative slices were originally selected. The resulting map per patient that you can observe at figure 8 can be overlaid on the original scan to indicate, in a form directly interpretable by a clinician, which anatomical regions the classifier’s decision rests upon. A **sanity metric** quantifying the fraction of Grad-CAM *energy falling outside the brain mask* was used to flag spurious activations concentrated on background or skull regions.

The second interpretability mechanism adopts a reconstruction-based approach using the group-specific autoencoders described in Section 4.1. During inference, each subject is reconstructed by all three class-specific autoencoders (CN, MCI, and AD), generating the anomaly maps shown in Figure 9. Regions with high reconstruction error are highlighted as anatomically atypical for the corresponding class, providing a complementary visualization of disease-related structural abnormalities.

Together, the two interpretability mechanisms offer converging but distinct forms of evidence: Grad-CAM explains a prediction by pointing to the regions that drove the classifier’s decision, while the anomaly maps show, independently of the classifier, where a subject’s anatomy departs from what is expected for a given diagnostic group.

Figure 8: GradCam result for subject *381_S_10511_sc*, diagnosed with Alzheimer’s disease.Figure 9: Autoencoder prediction for subject *005_S_10835_sc*, diagnosed with Alzheimer’s disease.

4.5 Innovative Aspects

The proposed framework contributes several design choices that, individually and in combination, distinguish it from existing state of the art solutions.

Per-class multimodal autoencoders for ternary staging. Instead of training a single autoencoder on healthy subjects for binary anomaly detection that cannot separate MCI from AD [8], the proposed framework employs three class-specific multimodal autoencoders (CN, MCI, AD).

Dual explainability with shared latent space. The use of a duality interpretability approach with three class-specific anomaly maps (reconstruction branch) and the four Grad-CAM heatmaps (ternary head plus three binary heads) is clinically valuable, as suggested in the literature [5].

Two-stage multimodal attention fusion. Different fusion strategies are adopted depending on the underlying architecture. In the autoencoder framework (Section 4.1), modality embeddings are combined through channel-wise concatenation to preserve all modality-specific information. In contrast, the hybrid CNN-based approach (Section 4.2) projects modality-specific features into a shared embedding space and fuses them using attention pooling, allowing the network to adaptively weight each MRI modality according to its relevance for the current subject while providing interpretable modality importance scores.

Anatomically-motivated Gaussian-weighted slice aggregation. Existing inter-slice aggregation methods typically rely on either uniform mean pooling [13] or learnable attention mechanisms [14, 15], while Gaussian weighting has previously been used only as an intra-slice spatial prior [16], where it is applied within a single MRI slice to emphasize central pixels over peripheral ones (e.g., to highlight a region of interest such as an organ or lesion), rather than to model the relative importance of *different slices* along the axial dimension.

5 Results

5.1 Autoencoder-Based Classification

The reconstruction-loss classifier, evaluated on 141 held-out sessions (75 CN, 54 MCI, 12 AD), reached a final weighted F1 of 0.52 after progressive refinement through bias calibration and a post-hoc MCI fallback rule (margin = 0.010).

Bias calibration shifts each decoder’s loss by a class-specific offset estimated on a stratified reference sample (e.g. 20 CN, 20 MCI, 49 AD), correcting for the systematic advantage that imbalanced training confers on minority-class decoders.

MCI fallback rule reassigns a session to MCI whenever the CN and AD calibrated losses are within a tunable margin of each other, reflecting the clinical premise that MCI occupies an anatomically intermediate position between the two. We optimized the margin by reviewing accuracy results and confusion matrix for different margin values. We choose a value of 0.010 as margin because it is a good trade off between the three classes.

Table 1 reports how each refinement step affects accuracy and per-class recall.

Configuration	Accuracy	CN Recall	MCI Recall	AD Recall
No calibration (raw loss)	0.150	0.090	0.040	1.000
Bias calibration (uniform sample)	0.400	0.520	0.240	0.420
Bias calibration (stratified sample)	0.496	0.710	0.220	0.420
Stratified calibration + MCI fallback ($m = 0.010$)	0.511	0.650	0.330	0.420

Table 1: Ablation over calibration strategies. Each row adds one refinement over the previous.

The final per-class performance of the best configuration is reported in Table 2.

Class	Precision	Recall	F1-score	Support
CN	0.670	0.650	0.660	75
MCI	0.450	0.330	0.380	54
AD	0.180	0.420	0.250	12
Weighted avg	0.540	0.510	0.520	141

Table 2: Classification report for the best configuration (stratified calibration + MCI fallback, margin = 0.010).

Loss distribution analysis. Figure 10 shows the MSE loss distributions produced by each group-specific decoder, coloured by true diagnostic group. Across all three decoders the loss ranges overlap substantially (≈ 0.016 – 0.032 MSE), with no decoder producing a clearly separated cluster for its own group.

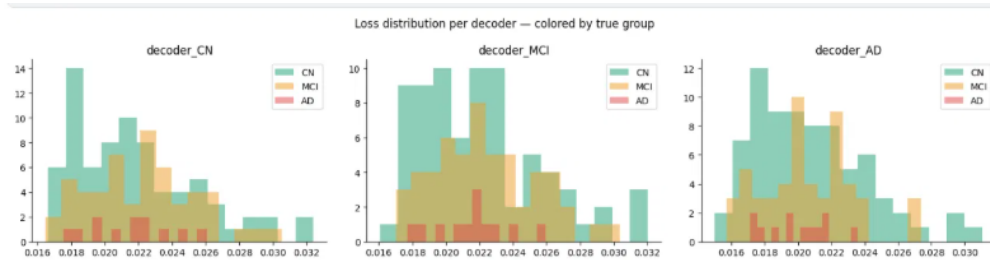


Figure 10: MSE loss distributions per decoder coloured by true diagnostic group (CN, MCI, AD).

Despite these gains, three structural limitations persist.

- **MCI recall remains low**, consistent with its clinical definition as a transitional state whose neuroanatomical profile overlaps substantially with both CN and AD.
- **AD precision is poor**: with only 12 test sessions the AD results are statistically fragile.
- **reconstruction loss provides a weak discriminative signal**: inter-decoder MSE differences are typically in the range $[0.001, 0.003]$, making the classifier highly sensitive to noise.

These results suggest that a purely reconstruction-based approach is insufficient for reliable Alzheimer’s staging, and motivate the adoption of supervised or contrastive objectives in future work.

5.2 Hybrid CNN based Classifier

In the three-class CN/MCI/AD task, the training accuracy stabilizes at approximately 0.423, which exactly matches the proportion of CN patients in the training set ($343/811 = 0.423$), while the balanced accuracy converges to 0.333 and the macro F1-score to 0.198. The same degenerate behavior is expected to generalize to the binary tasks, where each head is predicted to always output the majority class of its respective pair. This suggests that the patient-level embeddings produced by the shared, reconstruction-trained encoders lack sufficient discriminative information for the classification task.

5.3 ResNet-18 Classifier

The proposed framework was assessed at the patient level for all four classification tasks: the three-class CN/MCI/AD model and the three binary specialist models (CN vs. MCI, MCI vs. AD, and CN vs. AD). For each subject, the posterior probabilities obtained from the sampled axial slices were aggregated through probability averaging to produce a single patient-level prediction. Results are reported for two evaluation settings: the development evaluation split, used for model selection and validation (Table 3), and the final held-out test split, obtained after retraining the selected models on the combined training and evaluation data (Table 4).

Table 3: Patient-level performance of the 2.5D multimodal classifier on the **evaluation split** (train-only models).

Task	Acc.	Bal. Acc.	Macro Prec.	Macro Rec.	Macro F1	Macro Spec.	AUC (OvR)
CN-MCI-AD	0.560	0.468	0.461	0.468	0.463	0.746	0.724
CN-MCI	0.721	0.690	0.730	0.690	0.693	0.690	0.687
MCI-AD	0.545	0.593	0.555	0.593	0.499	0.593	0.552
CN-AD	0.759	0.650	0.597	0.650	0.607	0.650	0.687

Table 4: Patient-level performance on the **test split**, using final models retrained on the combined train+eval pool.

Task	Acc.	Bal. Acc.	Macro Prec.	Macro Rec.	Macro F1	Macro Spec.	AUC (OvR)
CN-MCI-AD	0.578	0.415	0.386	0.415	0.400	0.743	0.666
CN-MCI	0.607	0.591	0.621	0.591	0.574	0.591	0.669
MCI-AD	0.827	0.492	0.419	0.492	0.453	0.492	0.692
CN-AD	0.869	0.542	0.934	0.542	0.541	0.542	0.828

When assessing performance on the **evaluation set**, looking solely at overall accuracy can be misleading due to class imbalance. While the binary CN vs. AD task achieves the highest raw accuracy (0.759), a closer look at the class-balanced metrics reveals that the *CN vs. MCI task actually* yields the most robust performance. This indicates that the model learned more stable, balanced features for distinguishing healthy controls from early cognitive decline than for the more extreme stages.

The *MCI vs. AD task proves to be the most challenging* binary problem (balanced accuracy of 0.593), reflecting the subtle and heterogeneous structural changes characterizing the later transition into Alzheimer’s disease.

As expected, the *three-class classification* (CN-MCI-AD) represents the *most demanding scenario* overall, yielding the lowest balanced accuracy (0.468) and highlighting the difficulty of modeling the full disease continuum simultaneously.

The limitations highlighted by the balanced metrics are further confirmed when examining the **confusion matrices** (Figure 11), which reveal the specific misclassification patterns driving these scores.

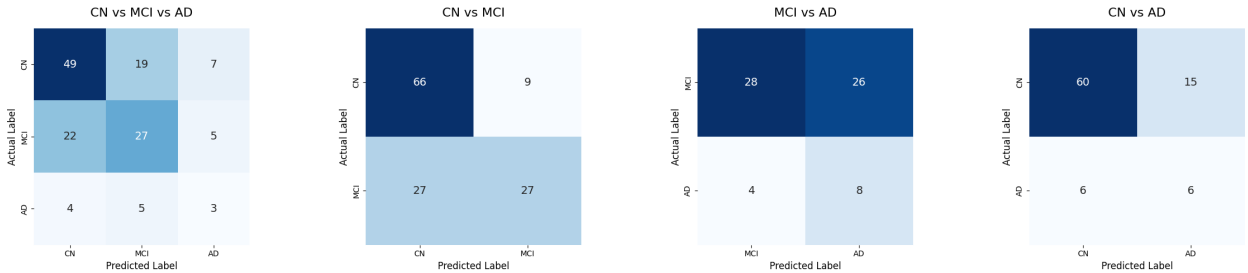


Figure 11: Confusion matrices for each task on the evaluation set

When evaluated on the official **test set**, the models exhibit a similar performance ranking, though with a moderate reduction in metrics. Overall, the test-set results confirm a critical insight: relying on raw accuracy heavily skews the interpretation of model success. When class imbalance is properly accounted for through balanced accuracy and F1 scores, the system demonstrates its strongest capability in differentiating cognitively normal subjects from those with Mild Cognitive Impairment.

6 Conclusion

6.1 Findings

This work developed and compared two complementary approaches to interpretable AD staging on ADNI4, using both reconstruction anomaly maps and Grad-CAM heatmaps for interpretability.

The initial design reported into Section 4.2 (Hybrid-CNN) collapsed to majority-class prediction across all four tasks, indicating that globally pooled patient-level embeddings from this architecture carried insufficient discriminative signal.

A purely reconstruction-based alternative (Section 4.1), improved substantially after bias calibration and a post-hoc MCI fallback rule, reaching a weighted F1 of 0.52, but inter-decoder loss differences remained small (0.001–0.003 MSE) and AD precision stayed poor given only 12 test sessions.

Replacing the classification branch with a supervised, transfer-learning 2.5D ResNet-18 classifier (Section ??), yielded the strongest results overall. The CN vs. AD task achieved the best separation (balanced accuracy 0.650, AUC 0.828 on the test split), consistent with the greater anatomical divergence between these two groups.

6.2 Strengths and Limitations

Strengths: Dual explainability and efficient design. The proposed framework’s main strength is its dual explainability, where agreement or divergence between Grad-CAM and reconstruction-based anomaly maps provides complementary anatomical evidence. Additionally, the ResNet-18 backbone ensures computational efficiency on limited hardware while leveraging transfer learning.

Limitation: Slice coverage. The current pipeline samples only 16 axial slices per volume (out of 128 after resampling) to remain within the memory constraints of Kaggle free tier. This constitutes a significant source of information loss: large portions of the brain volume, including regions known to exhibit AD-related atrophy, may fall between sampled slices and be entirely excluded from the embedding.

We attribute part of the gap between our validation performance and published state-of-the-art results to this constraint. Increasing the number of sampled slices to 32 or 64 is expected to yield meaningful performance improvements without any architectural change.

6.3 Future Work

Explicit Temporal Progression Modeling The ADNI-4 dataset contains multiple sessions per patient (screening, baseline, 12-month, 24-month), each with its own independent label. A natural extension is to exploit this longitudinal structure by treating each patient’s sessions as a time-ordered sequence and applying an LSTM to the sequence of session-level feature vectors produced by the shared encoder [12].

This is particularly relevant for MCI, which by clinical definition is a transitional stage lacking a stable, uniform anatomical phenotype: subjects labelled MCI may present anatomy anywhere along a continuum between CN and AD.

Patch-based processing. Splitting each MRI scan into a grid of small patches (e.g. 8×8) before feeding them to the nine encoders would, in principle, allow finer-grained, more spatially precise reconstruction and anomaly localization, since each patch could be reconstructed and scored independently rather than relying on a single whole-image embedding per modality. This would also make the anomaly maps more fine-grained, potentially highlighting small, localized anatomical changes that a whole-image reconstruction might average out. However, with nine encoders already active per forward pass in the current design, patch-based processing would multiply the number of forward passes by the number of patches per image, which is not feasible under the computational budget available (Kaggle free tier).

Alternative embedding aggregation strategies. The use of alternative aggregation strategies to the current Gaussian-weighted mean could improve the sensitivity of the session-level representation to localized anatomical patterns, particularly in regions such as the hippocampus that are critically affected in Alzheimer’s disease.

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